2026 CLIA Allowable Total Error (TEa) Reference Table

Proficiency testing acceptance limits under 42 CFR Part 493, Subpart I (CLIA PT final rule CMS-3355-F, published July 11, 2022, effective July 11, 2024). These limits grade proficiency testing.

Routine Chemistry (38)

Analyte	CLIA Allowable Error (TEa)	CFR
Alanine Aminotransferase (ALT/SGPT)	±15% or ±6 U/L (greater)	§493.931
Albumin	±8%	§493.931
Alkaline Phosphatase	±20%	§493.931
Amylase	±20%	§493.931
Aspartate Aminotransferase (AST/SGOT)	±15% or ±6 U/L (greater)	§493.931
Bilirubin, Total	±20% or ±0.4 mg/dL (greater)	§493.931
Blood Gas pCO2	±8% or ±5 mm Hg (greater)	§493.931
Blood Gas pO2	±15% or ±15 mmHg (greater)	§493.931
Blood Gas pH	±0.04	§493.931
B-Natriuretic Peptide (BNP)	±30%	§493.931
Pro B-Natriuretic Peptide (proBNP)	±30%	§493.931
Calcium, Total	±1.0 mg/dL	§493.931
Carbon Dioxide (CO2/Bicarbonate)	±20%	§493.931
Chloride	±5%	§493.931
Cholesterol, Total	±10%	§493.931
Cholesterol, HDL	±20% or ±6 mg/dL (greater)	§493.931
Cholesterol, LDL (direct)	±20%	§493.931
Creatine Kinase (CK)	±20%	§493.931
CK-MB Isoenzymes	±25% or ±3 ng/mL (greater) or MB elevated (presence or absence)	§493.931
Creatinine	±10% or ±0.2 mg/dL (greater)	§493.931
Ferritin	±20%	§493.931
Gamma Glutamyl Transferase (GGT)	±15% or ±5 U/L (greater)	§493.931
Glucose (excluding home use devices)	±8% or ±6 mg/dL (greater)	§493.931
Hemoglobin A1c (HbA1c)	±8%	§493.931
Iron, Total	±15%	§493.931
Lactate Dehydrogenase (LDH)	±15%	§493.931
Magnesium	±15%	§493.931
Phosphorus	±10% or ±0.3 mg/dL (greater)	§493.931
Potassium	±0.3 mmol/L	§493.931
Prostate Specific Antigen (PSA), Total	±20% or ±0.2 ng/mL (greater)	§493.931
Sodium	±4 mmol/L	§493.931
Total Iron Binding Capacity (TIBC)	±20%	§493.931
Total Protein	±8%	§493.931
Triglycerides	±15%	§493.931
Troponin I	±30% or ±0.9 ng/mL (greater)	§493.931
Troponin T	±30% or ±0.2 ng/mL (greater)	§493.931
Urea Nitrogen (BUN)	±9% or ±2 mg/dL (greater)	§493.931
Uric Acid	±10%	§493.931

General Immunology (20)

Analyte	CLIA Allowable Error (TEa)	CFR
Alpha-1 Antitrypsin	±20%	§493.927
Alpha-Fetoprotein (AFP, tumor marker)	±20%	§493.927
Antinuclear Antibody (ANA)	±2 dilutions or positive or negative	§493.927
Antistreptolysin O (ASO)	±2 dilutions or positive or negative	§493.927
Anti-HIV	Reactive (positive) or nonreactive (negative)	§493.927
Complement C3	±15%	§493.927
Complement C4	±20% or ±5 mg/dL (greater)	§493.927
C-Reactive Protein (hs-CRP)	±30% or ±1 mg/L (greater)	§493.927
HBsAg (Hepatitis B Surface Antigen)	Reactive (positive) or nonreactive (negative)	§493.927
Anti-HBc (Hepatitis B Core Antibody)	Reactive (positive) or nonreactive (negative)	§493.927
HBeAg (Hepatitis B e Antigen)	Reactive (positive) or nonreactive (negative)	§493.927
Anti-HBs (Hepatitis B Surface Antibody)	Reactive (positive) or nonreactive (negative)	§493.927
Anti-HCV (Hepatitis C Antibody)	Reactive (positive) or nonreactive (negative)	§493.927
IgA	±20%	§493.927
IgE	±20%	§493.927
IgG	±20%	§493.927
IgM	±20%	§493.927
Infectious Mononucleosis	±2 dilutions or positive or negative	§493.927
Rheumatoid Factor (RF)	±2 dilutions or positive or negative	§493.927
Rubella	±2 dilutions or positive or negative or immune or nonimmune	§493.927

Endocrinology (18)

Analyte	CLIA Allowable Error (TEa)	CFR
Cancer Antigen 125 (CA-125)	±20%	§493.933
Carcinoembryonic Antigen (CEA)	±15% or ±1 ng/mL (greater)	§493.933
Cortisol	±20%	§493.933
Estradiol	±30%	§493.933
Folate, Serum	±30% or ±1 ng/mL (greater)	§493.933
Follicle Stimulating Hormone (FSH)	±18% or ±2 IU/L (greater)	§493.933
Free Thyroxine (Free T4)	±15% or ±0.3 ng/dL (greater)	§493.933
Human Chorionic Gonadotropin (hCG)	±18% or ±3 mIU/mL (greater) or positive or negative	§493.933
Luteinizing Hormone (LH)	±20%	§493.933
Parathyroid Hormone (PTH)	±30%	§493.933
Progesterone	±25%	§493.933
Prolactin	±20%	§493.933
Testosterone	±30% or ±20 ng/dL (greater)	§493.933
T3 Uptake	±18%	§493.933
Triiodothyronine (T3)	±30%	§493.933
Thyroid Stimulating Hormone (TSH)	±20% or ±0.2 mIU/L (greater)	§493.933
Thyroxine (T4)	±20% or ±1.0 mcg/dL (greater)	§493.933
Vitamin B12	±25% or ±30 pg/mL (greater)	§493.933

Toxicology (14)

Analyte	CLIA Allowable Error (TEa)	CFR
Acetaminophen	±15% or ±3 mcg/mL (greater)	§493.937
Alcohol, Blood (Ethanol)	±20%	§493.937
Blood Lead	±10% or ±2 mcg/dL (greater)	§493.937
Carbamazepine	±20% or ±1.0 mcg/mL (greater)	§493.937
Digoxin	±15% or ±0.2 ng/mL (greater)	§493.937
Gentamicin	±25%	§493.937
Lithium	±15% or ±0.3 mmol/L (greater)	§493.937
Phenobarbital	±15% or ±2 mcg/mL (greater)	§493.937
Phenytoin (Dilantin)	±15% or ±2 mcg/mL (greater)	§493.937
Salicylate	±15% or ±2 mcg/mL (greater)	§493.937
Theophylline	±20%	§493.937
Tobramycin	±20%	§493.937
Valproic Acid, Total	±20%	§493.937
Vancomycin	±15% or ±2 mcg/mL (greater)	§493.937

Hematology (8)

Analyte	CLIA Allowable Error (TEa)	CFR
CBC - WBC (White Blood Cell Count)	±10%	§493.941
CBC - RBC (Red Blood Cell Count)	±4%	§493.941
CBC - Hemoglobin	±4%	§493.941
CBC - Hematocrit	±4%	§493.941
CBC - Platelet Count	±25%	§493.941
White Blood Cell Differential	±3 SD based on the percentage of different types of white blood cells in the samples	§493.941
Cell Identification	80% or greater consensus on identification	§493.941
Fibrinogen	±20%	§493.941

Coagulation (3)

Analyte	CLIA Allowable Error (TEa)	CFR
Prothrombin Time (PT)	±15%	§493.941
INR (International Normalized Ratio)	±15%	§493.941
Partial Thromboplastin Time (PTT/aPTT)	±15%	§493.941

Immunohematology (5)

Analyte	CLIA Allowable Error (TEa)	CFR
ABO Group	100% accuracy	§493.959
Rh Group (D typing)	100% accuracy	§493.959
Antibody Detection (Unexpected Antibodies)	100% accuracy	§493.959
Antibody Identification	80%+ accuracy	§493.959
Compatibility Testing (Crossmatch)	100% accuracy	§493.959

Source: 42 CFR Part 493, Subpart I. Provided by VeritaAssure® | Veritas Lab Services. Verify current requirements against the eCFR. Live lookup: www.veritaslabservices.com/resources/clia-tea-lookup.